Computer Networks — Ammar Ahmed

Assignment -4 — 20L-0961 — Date:

Review Questions

R18. Time-to-live (TTL) field

R19. No common bytes are used to compute the checksum in IP datagram
IP header checksum only computes checksum of IP packet's IP header fields which share no common bytes with IP datagram transport-layer segment part.

Problems:

P8

a)

longest prefix matching	link interface
11100000 00	0
11100000 01000000	1
11100000	2
11100000 1	3
otherwise	3

b) first address
prefix match 5th table entry; link interface 3
Second address
Prefix match 3rd table entry; link interface 2
third address
prefix matching 4th table entry; link interface 3

P9:

Destination Address Range	Link Interface
00000000 – 00111111	0
01000000 – 01011111	1
01100000 – 01111111	2
10000000 – 10111111	2
11000000 – 11111111	3

address count for interface 0 = $2^6 = 64$

address count for interface 1 = $2^5 = 32$

address count for interface 2 = $2^5 + 2^6 = 32 + 64 = 96$

address count for interface 3 = $2^6 = 24$.

P21

S2 flow table

Match	Action
Ingress Port = 1 ; IP Src = 10.3.*.* ; IP Dst = 10.1.*.*	fwd (2)
Ingress Port = 2 ; IP Src = 10.1.*.* ; IP Dst = 10.3.*.*	fwd (1)
Ingress Port = 1 ; IP Dst = 10.2.0.3	fwd (3)
Ingress Port = 2 ; IP Dst = 10.2.0.3	fwd (3)
Ingress Port = 1 ; IP Dst = 10.2.0.4	fwd (4)
Ingress Port = 2 ; IP Dst = 10.2.0.4	fwd (4)
Ingress Port = 4	fwd (3)
Ingress Port = 3	fwd (4)

Date:

Question 170.25.96.0/20 -> 4096 addresses

(I) Subnet mask

first group $2^8 = 256 - 1$ -> 255.255.255.0 /24

third group $2^4 = 16 - 1$ -> 255.255.255.240 /28

(II) net address of 1st customer

32-8 first group (1) 170.25.96.0/24 170.25.96.255 /24

8 customers ~~(2) 170.25.96~~ last address -> 170.25.103

32-6 Second group (2) 170.25.104.0 /26 170.25.107.63/26

16 customers last address -> 170.25.107.191/26

32-4 third group (1) 170.25.108.0 /28 170.25.108.16/28

64 customers last address -> 170.25.111

(III)

Broadcast address of last customer

first subnet (BC)

First group 170.25.103.0/24 -> 170.25.103.255/24

Second group 170.25.107.191 /26 170.25.107.255/26

Third group 170.25.111.240 /28 170.25.111.255/28